

Software Engineering

(UE24CS341A)

TEAM NUMBER: 007

TEAM MEMBERS:

KASISSNU SSINHA – PES1UG24AM130
SONIA SHARMA – PES1UG24AM904
P. SAANVI – PES1UG24AM905
KAUSTUBH AKASH – PES1UG24AM131

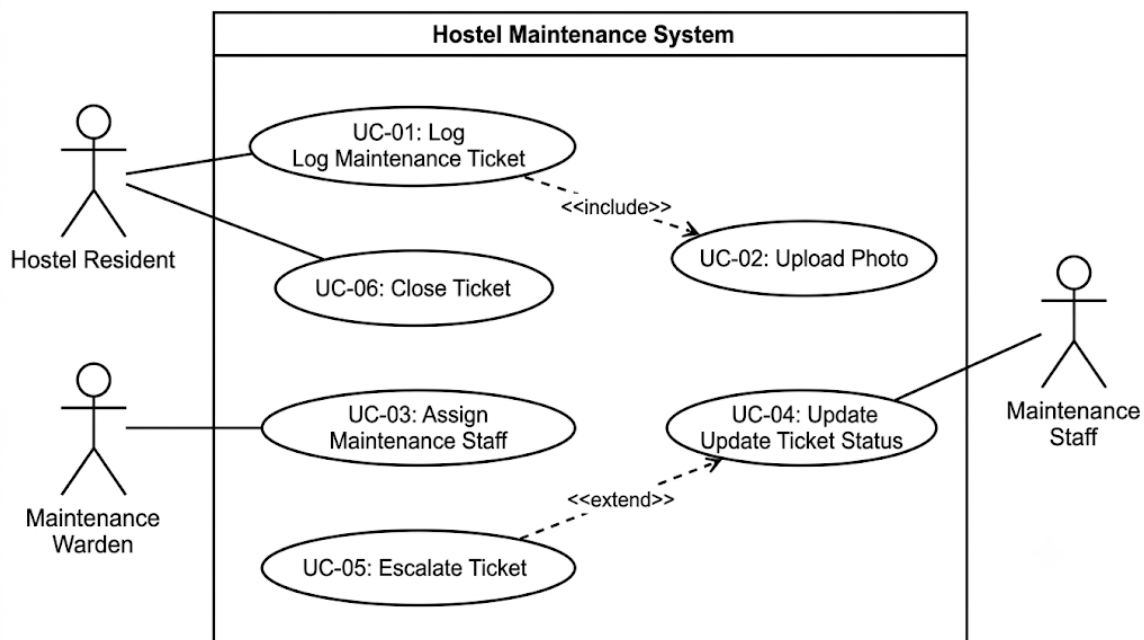
Lab 1: Hostel Maintenance & Issue Ticketing System

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: Ticket is created, assigned a unique ID, and routed to warden queue; Fail: Incomplete form accepted without room tag.	Core functionality for issue reporting
FR-002	Functional	The system shall allow the Maintenance Warden to assign an open ticket to a specific Maintenance Staff member.	High	Pass: Warden can select staff from a dropdown and ticket status changes to Assigned.	Necessary for dispatching staff
FR-003	Functional	The system shall allow Maintenance Staff to update the status of an assigned ticket (e.g., In Progress, Resolved).	High	Pass: Staff updates status to Resolved and timestamp is recorded.	Tracks progress of the issue
FR-004	Functional	The system shall require the Hostel Resident to digitally	Medium	Pass: Ticket status changes to Closed	Ensures resident satisfaction

		sign off to close a resolved ticket.		after resident confirms.	before closure
FR-005	Functional	The system shall display a dashboard of all active tickets for the Maintenance Warden, sortable by severity and time.	Medium	Pass: Warden sees a list of tickets sortable by severity.	Allows warden to oversee and prioritize work
NFR-001	Non-functional	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status change.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures urgent issues are not ignored
NFR-002	Non-functional	The system shall have an uptime of 99.9% during the academic semester.	Medium	Pass: Uptime logs show ≤ 43 minutes of downtime per month.	Residents must be able to report urgent issues anytime

2. UML Use-Case Diagram



3. Actors & Use Cases

Actors: Hostel Resident, Maintenance Warden, Maintenance Staff

Use-Case Flow Specification

Use Case: UC-06 Close Ticket

Primary Actor: Hostel Resident

Preconditions:

1. The resident is authenticated and logged into the system.
2. A maintenance ticket raised by the resident has been resolved by the Maintenance Staff.
3. The ticket status is **Resolved** and is awaiting resident confirmation.

Postconditions:

The resident's confirmation/sign-off is recorded, and the maintenance ticket status is changed to **Closed**.

Main Success Scenario:

1. Resident navigates to the **"My Maintenance Tickets"** section.
2. System displays the resident's maintenance tickets, including tickets with **Resolved** status.
3. Resident selects the resolved ticket they want to close.
4. System displays the ticket details and resolution information.
5. Resident reviews the completed maintenance work.
6. Resident confirms that the issue has been resolved and provides a **digital sign-off/confirmation**.
7. System records the resident's confirmation along with the date and time.
8. System changes the ticket status from **Resolved** to **Closed**.
9. System displays a success message confirming that the ticket has been closed.

Alternate Flow:

5a. Resident is not satisfied with the resolution

1. Resident selects the option indicating that the issue has **not been resolved**.
2. System displays an option for the resident to provide a reason or comment.
3. Resident submits the feedback.
4. System records the feedback and keeps the ticket open with its current **Resolved** status or reopens it for further action.
5. System notifies the Maintenance Warden that the resident has not confirmed the resolution.
6. Use case ends without closing the ticket.